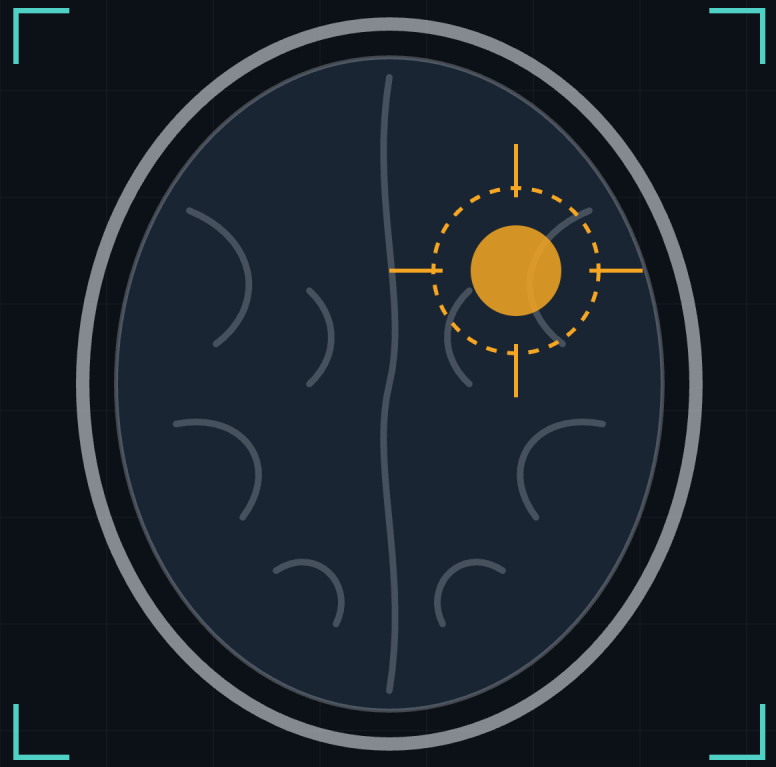
CV RESEARCH-REPRODUCTION PROJECT · M4–M5 REVIEW

Flatten or Pool?

One hypothesis: what we predicted, what happened, and why



Dally R · Kalash Kumari Thakur

Rishihood University

RECAP · M1-M3 IN ONE SLIDE

Where we left off

TASK

4 classes

glioma · meningioma · pituitary · no tumor

PAPER CLAIMS

99.24%

Wong et al. 2025, VGG16 transfer learning

WE REPRODUCED

95.25%

762 of 800 correct; glioma recall 84.5%

WE FOUND

202 / 800

test images repeat training images (25%)

PIPELINE

MRI → grayscale 224×224 → augment ×3 → VGG16 → Flatten → Dense 256 → 4 classes

THE GAP WE TARGETED · G4

The model memorizes

6.4 M weights in the Flatten head ÷ **5,712** training images (base paper) = **1,124** weights per image

training loss ≈ 0.02–0.03 vs validation loss ≈ 0.12 in the paper 1.33 in our reproduction

HYPOTHESIS H1 · STATED BEFORE RUNNING

Replace Flatten with Global Average Pooling: 49× fewer weights, so less to memorize, so better on new scans.

one variable | same data · same seed | 6.4 M → 0.13 M weights

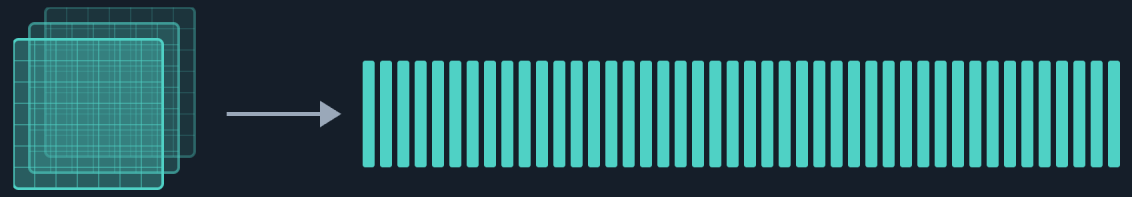
HYPOTHESIS H1

What actually changes

BASELINE

Flatten

keeps where every pattern is



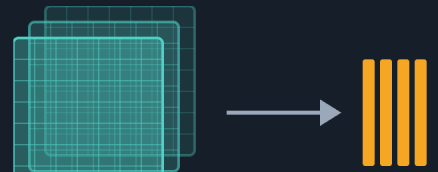
6.4 M

25,088 values in

OURS

Global Avg Pooling

keeps how strongly it appears



0.13 M

512 values in

VGG16 outputs 512 feature maps of 7 × 7. Flatten lists all 25,088 numbers; GAP averages each map to one.

PREDICTIONS · WRITTEN BEFORE RUNNING

Five predictions, one risk

TEST ACCURACY

+1 to +3 pts

less memorizing

VALIDATION LOSS

30–60% lower

fewer confident errors

TRAIN–VAL GAP

smaller

direct overfitting sign

GLIOMA RECALL

higher

weakest class

MACRO AUC

slightly higher

better ranking

RISK WE FLAGGED

pituitary may drop

GAP discards position

RESULTS · SAME CODE, SAME SEED, RUN TWICE

Flatten vs GAP, run twice

TEST ACCURACY — run 1 / run 2

+0.50 / −3.13

pts · 94.50 → 95.00% and 95.38 → 92.25%

GLIOMA RECALL — run 1 / run 2

+1.5 / −12.5

pts · 81.5 → 83.0% and 85.0 → 72.5%

UNSEEN IMAGES — run 1 / run 2

+0.67 / −4.18

pts · accuracy on the 598 non-leaked test images

VALIDATION LOSS — run 1 / run 2

−15% / +46%

0.667 → 0.565 and 0.616 → 0.899 · lower is better

MACRO AUC — run 1 / run 2

+0.003 / −0.003

0.989 → 0.993 and 0.989 → 0.986

STAGE-1 VAL LOSS — run 1 / run 2

−42% / −47%

GAP vs Flatten while VGG16 frozen: the effect that replicated

H1 PARTLY SUPPORTED — Same setup, opposite outcomes: GAP spread 2.75 pts vs 0.88 for Flatten (3 runs).

RESULTS · PREDICTIONS VS REALITY

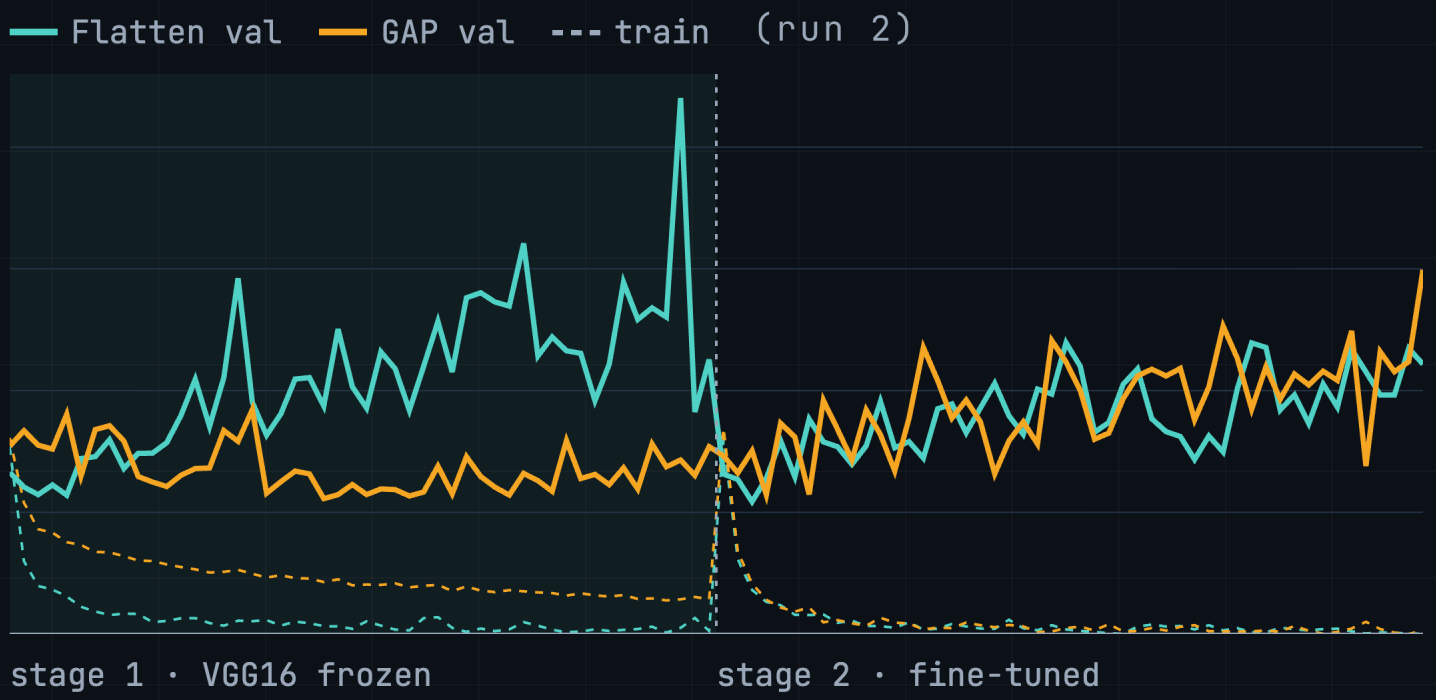
Predictions: mixed in run 1, failed in run 2

	MEASURE	PREDICTED	RUN 1	RUN 2
✕	Test accuracy	+1 to +3 pts	+0.50 pts	-3.13 pts
✕	Validation loss	30–60% lower	15% lower	46% higher
✕	Train–val gap	smaller	0.90 → 1.18	1.06 → 1.09
–	Glioma recall	higher	81.5 → 83.0%	85.0 → 72.5%
–	Macro AUC	slightly higher	0.989 → 0.993	0.989 → 0.986
–	Risk: pituitary	flagged: may drop	100 → 99.5%	99 → 100%

No prediction held in both runs. The one effect that replicated is in stage 1 (next slide).

WHY IT FAILED · THE KEY EVIDENCE

Overfitting comes from fine-tuning



STAGE 1

−42% / −47%

GAP validation loss vs Flatten while frozen, in run 1 / run 2. Replicated.

STAGE 2

≈ equal

Fine-tuning 13 M backbone weights made both overfit, in both runs.

WHY IT FAILED · POSSIBLE CAUSES

Six suspects, ranked by evidence

01 Fine-tuning overfits — STRONG

Stage 2: both models' val loss ≈ 1.0–1.1

02 GAP drops location — LIKELY

Stage-1 val acc −4 pts in both runs; glioma errors 30 → 55 in run 2

03 Fixed copies memorized — LIKELY

Train accuracy ≈ 100% on 16,800 fixed images

04 Last epoch kept — LIKELY

Val loss rises late; best epoch not kept

05 Messy data — PARTLY

Mixed sequences, a CT-like image, repeats

06 Run-to-run variance — STRONG

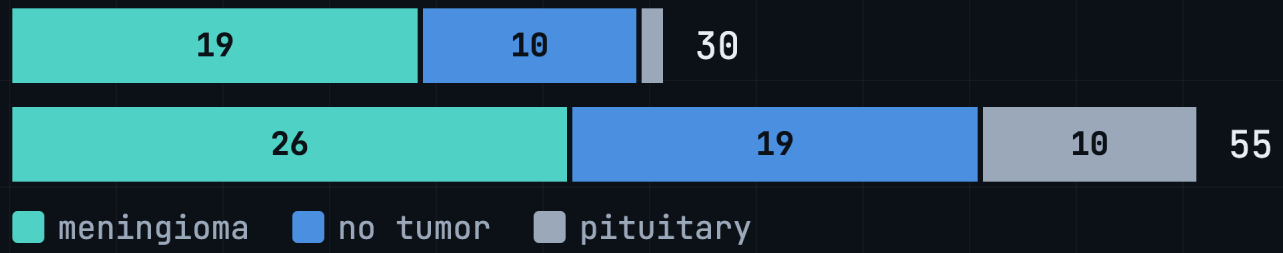
GAP: 95.00% vs 92.25% in two identical runs

Suspects 01, 03, 04 and 06 point to the same fix: fresh augmentations every epoch + best checkpoint + several seeds.

FAILURE ANALYSIS

Where it breaks: glioma

MISSED GLIOMAS (OF 200) AND WHAT THEY WERE CALLED · RUN 2



01

Confidently wrong

Worst errors made at ≈ 100% confidence.

02

Look-alikes

Large, sharply outlined gliomas resemble meningiomas.

03

Messy data

Mixed MRI sequences, a CT-like image, repeats.

TAKEAWAYS

What we learned

01

Reproduced

95.25% vs 99.24% claimed; 202 of 800 test images leak.

02

H1 partly supported

Stage-1 overfitting halved in both runs; final accuracy unstable (+0.50 / −3.13 pts).

03

Why

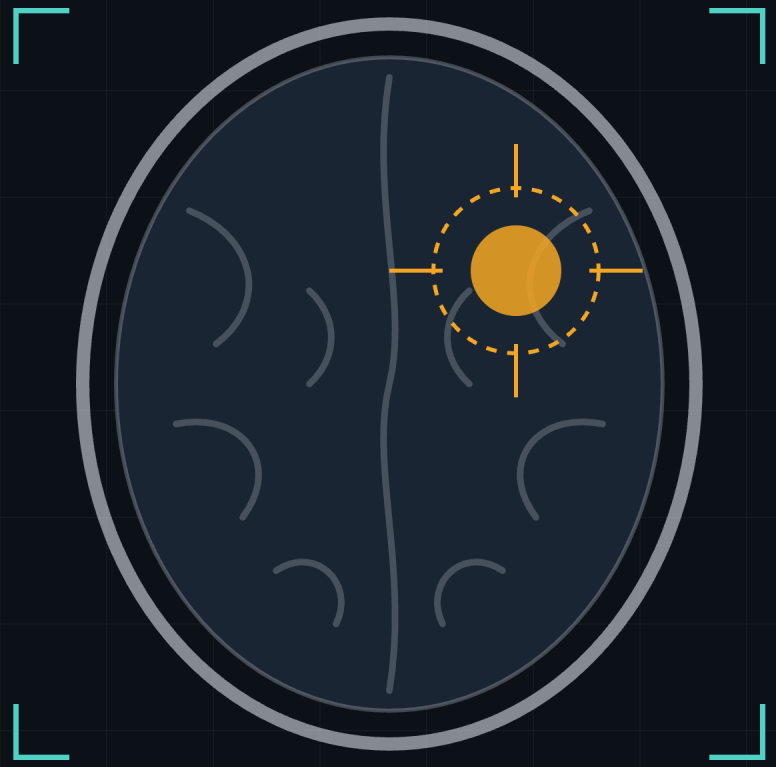
Fine-tuning, not the head, drives overfitting and makes results vary between runs.

NEXT Fresh augmentations every epoch + keep the best checkpoint + several seeds.

One dataset · image-level split · two ablation runs · no clinical claims

THANK YOU

Questions?



Dally R · Kalash Kumari Thakur

Base paper: Wong et al., PLoS ONE 20(5): e0322624, 2025